

Program: Diploma in Microelectronics	
Course Code: 5493C	Course Title: Programming with VHDL
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To explore VHDL and its capabilities
- To learn to use VHDL design units
- To build VHDL models using different modeling styles

Course Prerequisites:

Topic	Course code	Course Title	Semester
Combinational & Sequential circuits		Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Understand VHDL design units and language elements	15	Understanding
CO2	Develop digital circuits by applying dataflow modeling	14	Applying
CO3	Apply structural modeling to develop digital electronics circuits	14	Applying
CO4	Develop digital designs using behavioral modeling	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3				
CO2	3	3	3				
CO3	3	3	3				
CO4	3	3	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand VHDL design units and language elements		
M1.01	Explain the capabilities of VHDL	2	Understanding
M1.02	Discuss different design units in VHDL	5	Understanding
M1.03	Explain different data types in VHDL	5	Understanding
M1.04	Discuss the application of operators in VHDL	3	Understanding
Contents: Introduction VHDL - History, Capabilities, Hardware Abstraction, Basic Terminology - Entity Declaration, Architecture Body - Structural Style of Modeling, Dataflow Style of Modeling, Behavioral Style of Modeling, Configuration Declaration, Package Declaration, Package Body, Model Analysis, Simulation. Basic Language Elements Identifiers, Data Objects, Data Types - Subtypes, Scalar Types, Composite Types, Access Types, Incomplete Types, File Types, Operators - Logical Operators, Relational Operators, Adding Operators, Multiplying Operators, Miscellaneous Operators			
CO2	Develop digital circuits by applying dataflow modeling		
M2.01	Implement digital logic using concurrent signal assignment	3	Applying
M2.02	Develop a digital design using conditional and selected signal assignment statement	3	Applying
M2.03	Apply block statement and assertion statement in digital logic design	4	Applying

M2.04	Develop digital designs using dataflow modeling	4	Applying
	Series Test - I	1	
Contents: Dataflow Modeling Concurrent Signal Assignment Statement, Concurrent versus Sequential Signal Assignment, Delta Delay Revisited, Multiple Drivers, Conditional Signal Assignment Statement, Selected Signal Assignment Statement, Block Statement, Concurrent Assertion Statement Programming Examples : Logic gates, Half adder, Full adder, 4:1 Multiplexer, 2-bit Comparator			
CO3	Apply structural modeling to develop digital electronics circuits		
M3.01	Model a design using component declaration and component instantiation	4	Applying
M3.02	Understand resolution function	2	Understanding
M3.03	Design digital electronics circuits using structural modeling	8	Applying
Contents: Structural Modeling Component Declaration, Component Instantiation, Resolving Signal Values Programming Examples : Half adder using gates, Full adder using half adder, 4-bit Ripple Carry Adder, 4:1 multiplexer using 2:1 mux, D-FF, JK-FF, T-FF			
CO4	Develop digital designs using behavioral modeling		
M4.01	Model the procedural type behavior of an entity using process	2	Applying
M4.02	Implement various sequential statements in behavioral modeling	6	Applying
M4.03	Design digital circuits using behavioral modeling	7	Applying
	Series Test - I	1	
Contents: Behavioral Modeling Process Statement, Variable Assignment Statement, Signal Assignment Statement, Wait Statement, If Statement, Case Statement, Null Statement, Loop Statement, Exit Statement, Next Statement, Assertion Statement, Other Sequential Statements, Multiple Processes. Programming Examples : 8:1 Multiplexer, D-FF, JK FF, T FF, 3-bit upcounter, 3-bit downcounter			

Text /Reference:

T/R	Book Title/Author
T1	VHDL Primer Third edition by J. Bhaskar, PHI
R1	Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, 2/e, TMH, 2008
R2	Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008.
R3	Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008
R4	Pedroni V. A., Circuit design with VHDL, PHI, 2008.
R5	Kevin Skahill, VHDL for Programmable Logic, Addison & Wesley, 1996

Online Resources:

Sl. No.	Website Link
1	https://www.edutechlearners.com/download/books/A%20VHDL%20Primer%20-%20Jayaram%20Bhasker.pdf
2	https://www.intel.com/content/www/us/en/programmable/customertraining/webex/VHDL/presentation_html5.html
3	https://books.google.co.in/books?id=KQvx_b0C9BIC&printsec=frontcover#v=onepage&q&f=false
4	https://www.youtube.com/@synthesisofdigitalsystems-5267/videos
5	https://www.udemy.com/course/vhdl-programming-for-beginners/